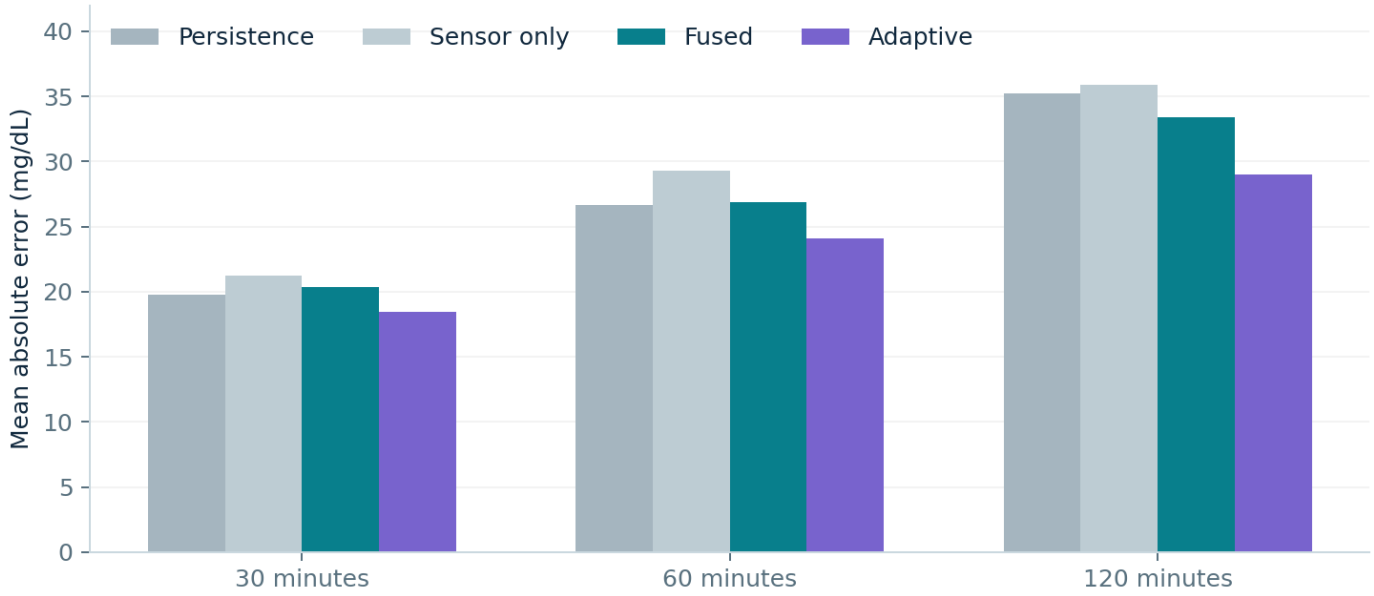
01 / GlucoTwin India

An adaptive partial Digital Twin for glucose forecasting

Digital Twin Challenge 2026 | Happiest Health
Dipanshu Raj | Team product | Indian Institute of Technology Kanpur

GlucoTwin combines the same participant's baseline clinical profile with glucose, wearable activity and recorded meals. It maintains an evolving patient state, produces direct forecasts at three horizons, and displays uncertainty and data-quality decisions in a working clinician research dashboard.

Primary outcome	Measured result	Interpretation
60-minute MAE	24.12 mg/dL	9.6% lower than persistence
120-minute MAE	29.00 mg/dL	17.7% lower than persistence
60-minute interval coverage	89.8%	Nominal 90%; empirical test coverage



Patient-macro errors on three held-out T2D participants; all methods use the same test set. The fused model alone does not outperform persistence at every horizon.

What the prototype establishes

The package demonstrates paired-stream fusion, causal state updates, delayed personalization and a functional review interface. Twelve core tests and desktop/mobile browser checks pass. It runs locally after setup without a paid API or GPU.

What it does not establish

Event alerts remain exploratory: 14 of 64 alerts match an upcoming onset, with 15 minutes median matched lead time. The small US cohort, upstream interpolation and lack of prospective Indian validation prevent claims of clinical readiness or treatment benefit.

02 / Clinical question and data

A narrow, measurable use case

The intended user is a clinician reviewing an adult with Type 2 diabetes between consultations. The primary research task is glucose prediction 60 minutes from issue time; 30 and 120 minutes are secondary horizons. Dynamic inputs are deliberately delayed by 15 minutes. A horizon is not a demonstrated warning lead time.

Exploratory event definition

A new onset occurs when glucose crosses above 180 mg/dL and remains above it for three consecutive five-minute bins. All three bins must lie within the next 60 minutes. This persistence rule is a project endpoint, not an emergency diagnosis or a patient-specific treatment target. The 180 mg/dL boundary is grounded in CGM reporting conventions [3].

Source stream	Fields actually used	Representation
Static / clinical	Age, sex, BMI, HbA1c, fasting glucose	EHR-like JSON; same participant ID
Dynamic / sensors	Dexcom glucose, heart rate, METs	Five-minute aligned time series
Recorded context	Meal carbohydrate, protein and fat	Causal decaying meal-state proxies

Dataset and cohort

CGMacros v1.0.0 contains 45 participants, including 14 in its Type 2 diabetes group, over approximately ten days per participant [1,2]. The study was conducted in the United States. The project retains participant identity across streams; it does not pair unrelated synthetic EHRs with real glucose data. Missing medication, genetic and diagnosis-history fields are explicitly marked as unavailable.

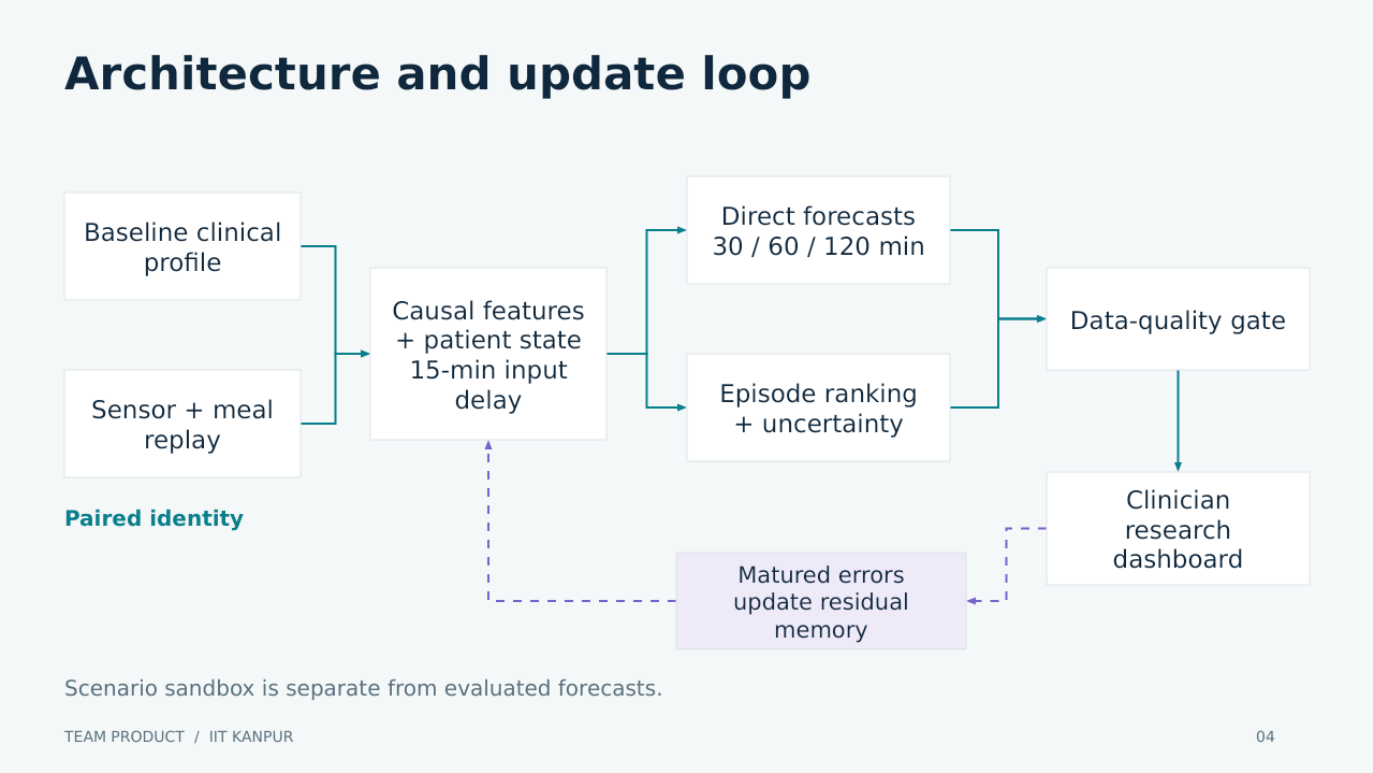
Conservative preprocessing

The published glucose stream is already interpolated to one-minute rows. Preparation retains values aligned with the initial five-minute Dexcom phase and integer-valued readings, then delays all dynamic inputs by 15 minutes. No additional backward interpolation is applied. One missing input bin may be carried forward; target values remain unfilled. Activity METs are divided by ten according to the source dictionary. Median retained target coverage is approximately 90.1%; individual coverage varies substantially.

Provenance and licensing

The original participant CSV subset, baseline table, per-file SHA-256 hashes and processed records are included. Data and derivatives retain CC BY-NC-SA 4.0. Original code is MIT licensed. Model distribution uses conservative noncommercial attribution/share-alike terms. No endorsement or commercial licensing permission is implied.

03 / Architecture and patient state



Direct forecasting

A histogram gradient-boosting regressor predicts change from the latest available glucose for each horizon. The sensor baseline uses lagged glucose, slopes, rolling statistics, time of day, heart rate, activity and missingness. Fusion adds recorded meal states and five baseline clinical features. Training uses 180 boosting iterations, 15 leaves, learning rate 0.06 and regularization; full parameters are in train.py.

Patient-specific feedback

For a horizon h, a forecast error is available only after h + 15 minutes. The residual is clipped to [-60, 60] mg/dL, then updates an exponentially weighted state: s(t) = 0.97 s(t-1) + 0.03 error(t). The adaptive correction is 0.35 s(t). This bounded memory personalizes the forecast without reading future outcomes. Validation selects the fused or adaptive variant separately at each horizon.

Interpretability boundary

The evolving state is inspectable, but the system is a data-driven partial twin, not a validated physiological replica. Its decaying meal states are feature proxies. The separate scenario model uses assumed response coefficients and cannot establish a causal treatment effect. There are no dosing or medication recommendations.

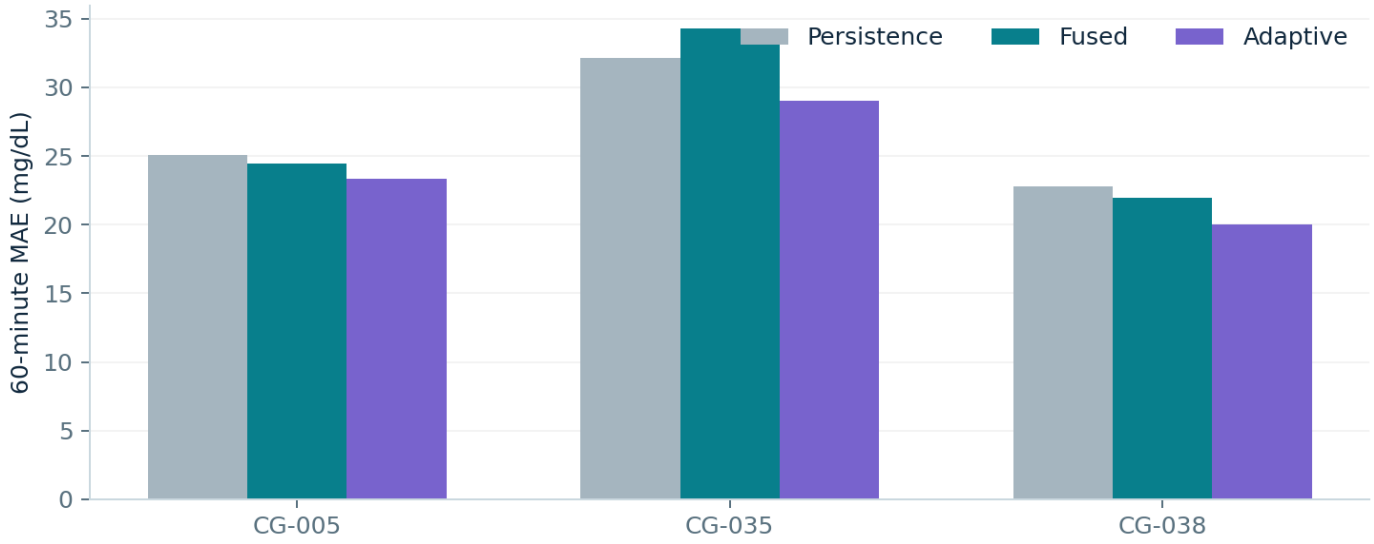
04 / Forecast evaluation

Partition	Participants	Purpose
Training	36	Fit models; includes other metabolic groups
Validation	3 T2D: 12, 46, 14	Choose variants and alert threshold
Calibration	3 T2D: 39, 47, 3	Estimate interval widths
Test	3 T2D: 35, 5, 38	Report final held-out performance

Split seed: 2026. The first 24 hours are warm-up. Training windows are sampled every 15 minutes and weighted equally by participant. Test windows are scored every five minutes. Participant-disjoint partitions prevent overlapping windows from crossing partitions. Headline errors are patient-macro means.

Horizon	Persistence	Sensor only	Fused	Adaptive
30 min	19.74	21.22	20.34	18.48
60 min	26.67	29.26	26.88	24.12
120 min	35.26	35.86	33.41	29.00

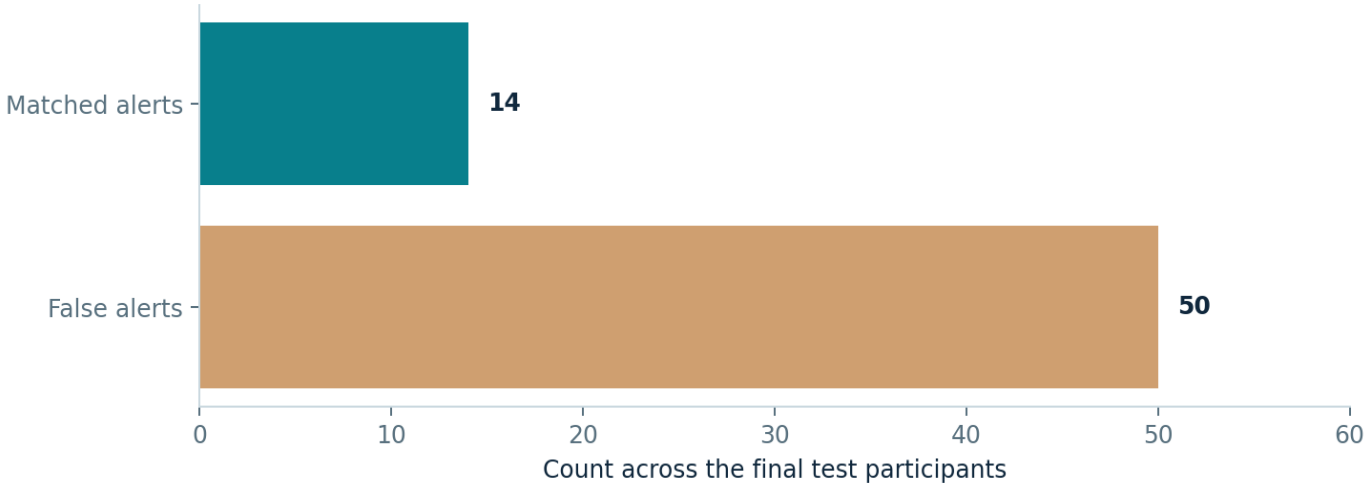
Mean absolute error, mg/dL. All adaptive variants were selected using validation participants before final testing. Per-patient RMSE and window counts are also included.



Uncertainty in the evidence

The 60-minute mean MAE improvement over persistence is 2.56 mg/dL. A patient-cluster bootstrap gives a descriptive 95% interval of 1.70 to 3.15 mg/dL, but it uses only three test people and is not robust population evidence. A larger external cohort and repeated, preregistered validation are required. The results do not establish that clinical-feature fusion alone improves every horizon.

05 / Alerts and forecast intervals



Alert measure	Measured value
Matched / issued alerts	14 / 64
Detected / observable onsets	14 / 66
Episode precision / recall	21.9% / 21.2%
False alerts / eligible patient-day	1.87
Median matched warning lead time	15 minutes
Eligible monitored time	26.78 patient-days

Each alert can match at most one onset; alerts use a 60-minute cooldown. An observable onset has an eligible earlier warning opportunity. False alerts use eligible monitored time as their denominator. The classifier is class balanced; its ranking score is not a calibrated clinical probability.

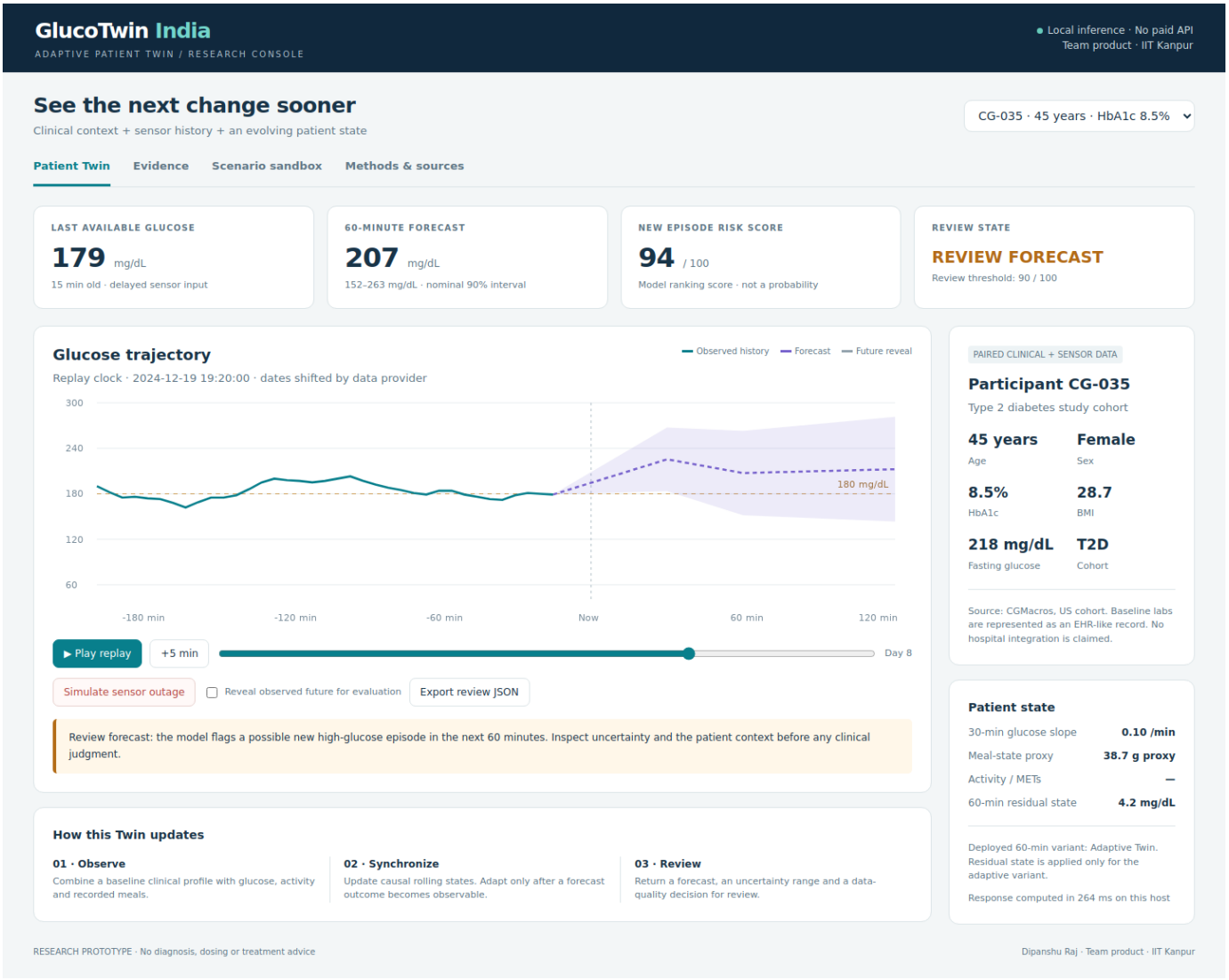
These alarms are not clinically deployable

The event module is included as an auditable exploratory experiment. Its false positives and missed episodes are material limitations. The primary performance claim concerns glucose forecast error, not reliable two-hour event warning or improved health outcomes.

Forecast horizon	Nominal interval	Patient-macro test coverage
30 minutes	90%	87.8%
60 minutes	90%	89.8%
120 minutes	90%	92.1%

Intervals use absolute forecast residuals from separate calibration participants. Temporal dependence and population shift mean nominal coverage is not a prospective or distribution-free guarantee. Wider intervals should be interpreted as uncertainty, not evidence of a precise forecast.

06 / Working clinician research console



Interaction	Implemented behavior
Patient replay	Select held-out participants; advance by five-minute steps
Forecast review	Display 30/60/120-minute values and residual intervals
Data-quality test	Simulated outage suppresses forecast and risk score
Evidence view	Show every baseline, patient scope and alert limitations
Scenario / export	Clearly synthetic assumptions; export review context as JSON

The displayed stream is accelerated playback of historical data. It is not a live wearable connection. Observed future is hidden unless the evaluator explicitly enables retrospective reveal. The local API exposes bundled test participants; general clinical data upload and hospital integration are not implemented.

07 / Engineering and reproducibility

Compact local architecture

The application uses Python 3.12, NumPy, pandas, SciPy, scikit-learn and joblib. A standard-library HTTP server binds to 127.0.0.1. The frontend uses HTML/CSS/JavaScript and Canvas with no remote scripts. After dependency setup, the packaged model and data run without internet, a GPU or a paid model API. This is not a hardened multi-user clinical deployment.

Check group	Verification
Temporal integrity	Prefix-invariant features; future perturbation does not alter the past
Feedback integrity	Residual updates wait until the target becomes observable
Model consistency	Prefix recomputation matches whole-trace causal predictions
Event labels	New onset required; two-bin spikes rejected; missing targets unscored
Quality and inputs	Outage suppresses outputs; invalid participant/row rejected
Interface	Replay, tabs, switching, reveal, export control and mobile layout

Reproduction commands

```
python -m pip install -r requirements.txt
python -m glucotwin.prepare
python -m glucotwin.train
python -m unittest discover -s tests -v
python app.py
```

Measured build and delivery

Training took 26.8 seconds on the build host with two OpenMP threads. This is hardware dependent. Twelve core unit tests pass. Browser checks report no page errors and no horizontal overflow at the tested mobile viewport. The package includes source-data checksums, the exact split, model artifact, per-patient results and a JSON evaluation summary.

Security and scope

The package uses open anonymized data and collects no new patient records. Arbitrary uploaded joblib files are not accepted. The trusted bundled artifact should be preserved; pickle-based model files are executable and should not be loaded from unknown sources. Public deployment, authentication, institutional governance and regulatory assessment are future work. No compliance certification is claimed.

08 / Limits, next steps and references

Limitation	Required next step
Three final test patients; US cohort	Larger, representative Indian external validation
Upstream glucose interpolation	Obtain native device exports and timestamp provenance
Unknown meal logging latency	Prospective capture with actual availability timestamps
Weak event precision and recall	Improve ranking, calibrate risk and test alarm burden
No intervention evidence	Independent physiological/causal validation before treatment simulation
No prospective clinical use	Clinician-supervised silent evaluation before visible alerts

India deployment pathway

The proposed next stage is a clinic-supervised research pilot using appropriately governed paired data. It should preregister endpoints, assess sensor and demographic subgroups, document missingness, and evaluate alert burden without changing treatment. Local execution reduces software service dependencies but does not establish CGM affordability, cost savings or access for all patients.

Submission ownership

Team product | Dipanshu Raj | B.Tech., Materials Science and Engineering | Indian Institute of Technology Kanpur. GitHub: Dipanshur19. No incubator affiliation is claimed. AI assistance was used for implementation and documentation; the participant should review and be able to defend the work.

Primary sources and evaluation artifacts

[1] Gutierrez-Osuna et al. CGMacros v1.0.0. PhysioNet (2025). https://doi.org/10.13026/3z8q-x658

[2] Das et al. CGMacros: a pilot scientific dataset for personalized nutrition and diet monitoring. Scientific Data (2025). https://www.nature.com/articles/s41597-025-05851-7

[3] Battelino et al. Clinical Targets for Continuous Glucose Monitoring Data Interpretation. Diabetes Care (2019). https://pubmed.ncbi.nlm.nih.gov/31177185/

[4] scikit-learn. Histogram-based gradient boosting documentation. https://scikit-learn.org/stable/modules/ensemble.html#histogram-based-gradient-boosting

[5] Project evidence: results/summary.json, forecast_metrics.csv, per_patient_metrics.csv; data/audit.json and split.json; tests/test_core.py.

[6] Data license: https://creativecommons.org/licenses/by-nc-sa/4.0/ . Original code: MIT; model/data terms are separate.